Re-Bottleneck: Latent Re-Structuring for Neural Audio Autoencoders

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Motivation

- Neural audio autoencoders (AEs) power audio processing.
- **Disconnect**
  Reconstruction-based AEs ↔ Downstream task needs.
- **Existing solutions**
  1. Adapt downstream models.
  2. Redesign and retrain the AE.

Can we impose desired latent structure on existing AEs, without costly retraining?

Re-Bottleneck Method

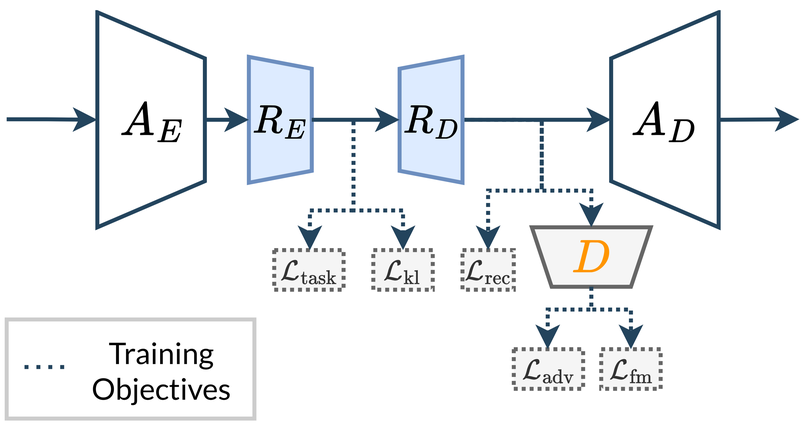
- Train a compact **inner autoencoder** (R_E, R_D), inside a base pre-trained and frozen VAE (A_E, A_D).
  - Training with **latent-domain losses** only.

Latent Space Training Objectives

- Latent reconstruction loss:

$$L^R_{\text{rec}} = \mathbb{E}_{\boldsymbol{x}} \left[\left\| A_E(\boldsymbol{x}) - R(A_E(\boldsymbol{x})) \right\|_2 \right].$$

- Latent **discriminator** (adversarial & feature matching loss).
- Task-specific **latent-structure loss**.
- KL-regularization.



Code:
github.com/dbralios/rebottleneck

Setup

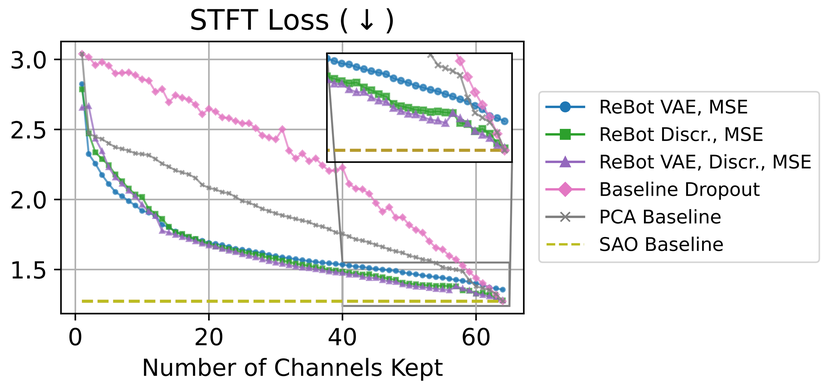
- **Base AE**: Stable Audio Open (SAO) VAE (156M params).
  - Stereo 44.1 kHz→ 64 channels at 21.5 Hz.
- **Inner autoencoder** R (19M params).
  - Training: 0.33% base AE GPU hrs.
- Latent diffusion experiments: SAO text-to-audio pipeline.

Experiments & Results

1. Ordered Re-Bottleneck

Hierarchical channel ordering.

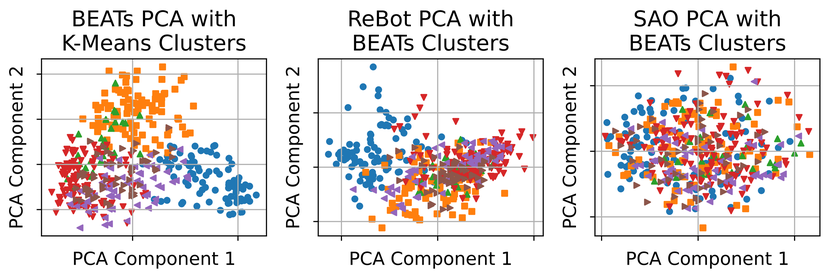
- Nested dropout training.
  - Sample length m, make mask $M^{(m)}_{c,t} = 1$ if $c \leq m$ else 0.
  - Apply element-wise $R^{(m)}(\boldsymbol{x}) = R_D(M^{(m)} \odot R_E(\boldsymbol{x}))$.



2. Semantically-Aligned Re-Bottleneck

- Alignment with semantic embeddings (BEATs, T5).
  - Linear probe + time-pooling → $\bar{z}$,
    Pooled semantic embeds → $\bar{s}$.

$$L^R_{\text{task}} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp\left(\text{sim}(\bar{z}_i, \bar{s}_i)/\tau\right)}{\sum_{j=1}^{B} \exp\left(\text{sim}(\bar{z}_i, \bar{s}_j)/\tau\right)}$$

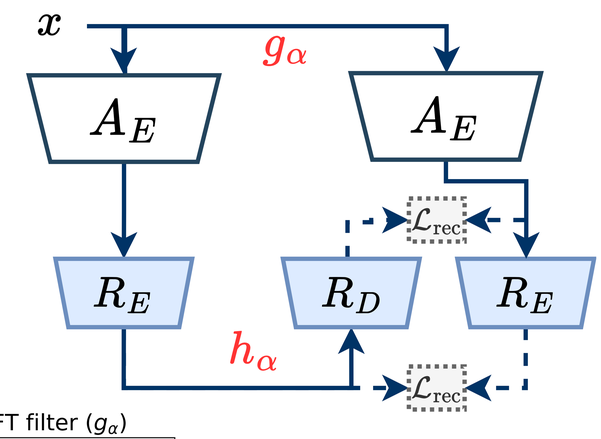


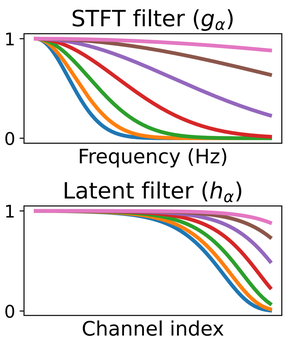
3. Equivariant Re-Bottleneck

Audio transformation $g_\alpha(\cdot)$ → Latent transformation $h_\alpha(\cdot)$.

- Low-pass filter → Multiplicative latent filter.

- Explicit inner bottleneck & implicit outer bottleneck loss.





STFT (↓) distance to low-pass filtered target.

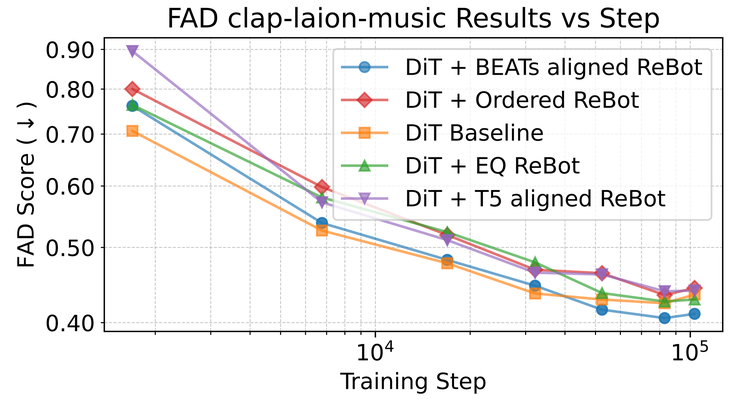
Cutoff (kHz)	ReBot	AE Target	SAO + Latent Filtering
1.4	0.98	0.88	5.08
2.8	1.08	1.05	4.10
5.5	1.17	1.16	2.60

Evaluating Reconstruction and Generation

Reconstruction performance

Model	STFT (↓)
T5 aligned ReBot	1.29
BEATs aligned ReBot	1.29
Equivariant ReBot	1.39
SAO Baseline	1.27

- Desired latent structure, preserved reconstruction.
- We train text-to-audio **diffusion models** on our structured latents.



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